



×



CLICK-A-THON 2026 · BENGALURU · SONYLIV TRACK

# Counting real viewers, not ghosts.

BACKGROUND-ONLY CONCURRENCY AT STREAMING SCALE

Count only what is **actually being watched** — exact peaks, corrections that are arithmetic instead of rebuilds, and verification that fails loud. Every number in this deck is from the **unseen day**, measured on the live service.

● **fastandfurious.live** — the concurrency curve, live

● **chat.fastandfurious.live** — ask it in plain English

| ENGINE                                          | UNSEEN DAY                          | PEAK                                     |
|-------------------------------------------------|-------------------------------------|------------------------------------------|
| ClickHouse Cloud 26.2 · ap-south-1 · 2 replicas | 7,000,000 events · 108,486 sessions | 19,882 concurrent · 2026-07-31 11:15 UTC |

# A session boundary is not a viewer.

Counting every session between its first and last event is the obvious implementation, and on the unseen day it is wrong by a fifth. Phones get pocketed, players get paused, streams stall — all of it keeps a session technically open and none of it is an audience.

NAIVE · SESSION BOUNDARIES

## 24,069

peak concurrent sessions, 2026-07-31 11:16 UTC.  
1,453,136 aggregate session-minutes.

MEASURED · FOREGROUND-ONLY

## 19,882

peak concurrent sessions, 11:15 UTC — a different minute.  
1,117,802 aggregate session-minutes.

OVERCOUNT AT PEAK

## +21.1%

Peak-vs-peak. The two peaks do not land on the same minute, so no correction factor fixes it after the fact.

OVERCOUNT IN AGGREGATE

## +30.0%

335,334 session-minutes of audience that was never watching anything.

AT THE SCALE IT MATTERS

## 4,187

phantom viewers at the single busiest minute of the day — the number ad load and capacity are sized against.

# Five questions.

## Five answers.

|   |                                                                                                                    |                                                                                                                                                                                                         |                                                   |
|---|--------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| 1 | Defining an <b>active interval</b> when the heartbeat is missing, the player is paused, or the app is backgrounded | A five-clause predicate in a versioned <code>policy.yaml</code> — <b>started</b> $\wedge$ $\neg$ <b>terminated</b> $\wedge$ <b>foreground</b> $\wedge$ <b>playing</b> $\wedge$ <b>live within 120 s</b> | <b>0.2%</b><br>gaps falsely split                 |
| 2 | Representing active ranges — arrays, intervals, minute deltas, or a hybrid                                         | <b>Hybrid.</b> Normalized intervals are the truth, signed $\pm 1$ deltas are the serving form, a dense minute tier sits on top                                                                          | <b>43x</b><br>fewer rows than events              |
| 3 | Minute-wise <b>peak and average</b> without scanning raw session history                                           | Prefix-sum from a day anchor, never $t = 0$ . Average is <b>time-weighted</b> , not sampled at boundaries                                                                                               | <b>5 ms</b><br>8,192 rows read                    |
| 4 | Staying <b>filter-friendly</b> across platform, country, content, video type, grain                                | Nine dimension combinations in the sort key. Peak computed <b>after</b> aggregating — never by summing per-dimension peaks                                                                              | <b>1,100 @ 11:23</b><br>a slice's own peak minute |
| 5 | Handling sessions that are <b>still open</b> and keep growing                                                      | A quiet session falls out of a <b>WHERE</b> clause — nothing is written when a viewer stops. History takes signed corrections                                                                           | <b>37,649</b><br>never closed. absorbed           |

# Seven million events in. 163,740 intervals out.

LANDING

events\_raw

7,000,000 rows · 41.4 MiB · 0 rejected

FAST BECAUSE

Sorted by session, then time — the only hot read it has. No dashboard ever reaches this table.

WE DID

Append-only, so the duplicate rate stays measurable. All three dedup windows pinned — Cloud silently moves them onto a timer.

NORMALISE

events\_clean

6,974,862 rows · 46.1 MiB · 25,138 collapsed

FAST BECAUSE

Read through an argMax view — correct whether or not a merge has run, and no FINAL on the hot path.

WE DID

Normalise once, at the boundary, so no GROUP BY downstream double-counts a casing variant.

ENRICH

content\_dict

33,326 titles · hashed, in memory

FAST BECAUSE

The metadata join happens once during compaction, not per query. Video type is denormalized into the interval rows.

WE DID

Assert enrichment resolved before writing, and force the load — a dictionary loads per replica, and an empty one still reports LOADED.

STAGE 01

active\_intervals

163,740 intervals · 96,844 sessions · 7.2 MiB

FAST BECAUSE

43× fewer rows than events, and an aggregate projection answers the revision subquery so the current-state view prunes.

WE DID

One 13-CTE state machine, in one file, for both the full rebuild and the incremental run — so the oracle cannot drift from what it validates.

# Two questions. Two structures.

STAGE 02

concurrency\_deltas

6,435,460 rows · 34.9 MiB · nine masks

FAST BECAUSE

Two rows per interval per mask, and the answer is `sum(delta)` over the full key — merges can lag without changing it.

WE DID

An explicit column list, because a bare Summing table sums your *dimensions*. Plus a conservation gate that throws with the ratio.

CHECKPOINT

bucket\_net · day\_anchor

1,004,654 + 190,980,048 rows · 32.8 MiB

FAST BECAUSE

A window read never sums from `t = 0`. It starts at the nearest anchor, so cost is bounded by the window, not by history.

WE DID

Anchored per service day, so each day is self-contained and a new day loads as an isolated partition.

SERVING

concurrency\_minute\_versions

1,295,876 rows · 4.3 MiB · dense minutes

FAST BECAUSE

No cumulative sum at read time at all. A whole-day peak *and* time-weighted average comes back in **5 ms reading 8,192 rows**.

WE DID

Immutable generations bound to a policy and a sealed snapshot. Peak computed **after** aggregating the requested combination.

LIVE, INDEPENDENT

session\_live\_now

96,844 session states · 2.6 MiB

FAST BECAUSE

O(open sessions), never history — identical cost on day 1 and day 300. A quiet session **falls out of the WHERE clause**.

WE DID

Nothing is written when a viewer stops, which is the whole hard problem: ClickHouse has no primitive that fires on absence.

# Two rows per interval.

## Here is what that buys.

| EVENTS IN                                        | ROWS WE KEEP                                                           | ROWS A DASHBOARD READS                                        | TIME TO ANSWER                                     | REBUILDS TO CORRECT IT                                   |
|--------------------------------------------------|------------------------------------------------------------------------|---------------------------------------------------------------|----------------------------------------------------|----------------------------------------------------------|
| <b>7,000,000</b><br>the unseen day, ingested raw | <b>163,740</b><br>43× fewer — bounded by sessions, not session-minutes | <b>8,192</b><br>1,700× fewer than recomputing the same answer | <b>5 ms</b><br>95× faster than the 476 ms raw scan | <b>0</b><br>a correction is subtraction, not a recompute |

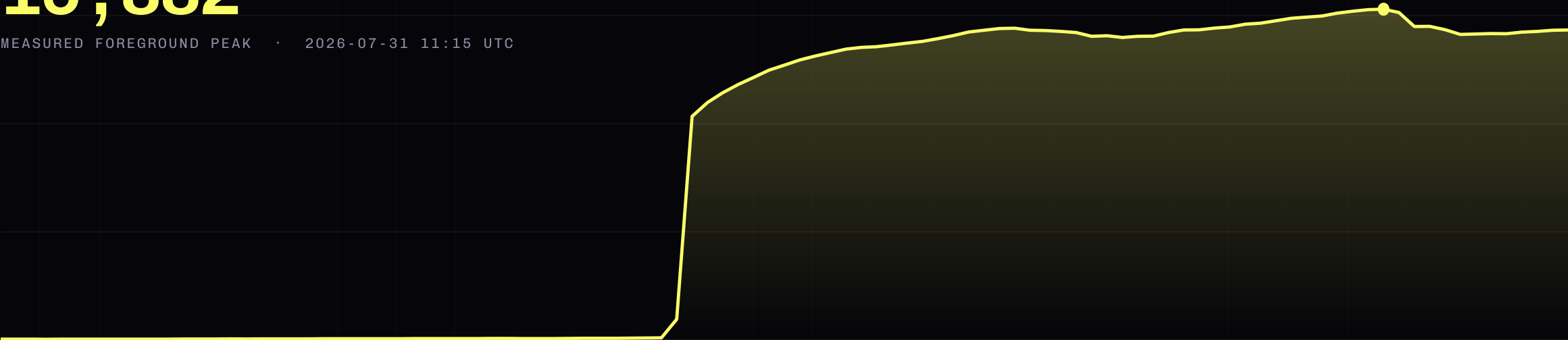


# The unseen day, answered.

24,069 naive session-boundary peak — what session boundaries claim

19,882

MEASURED FOREGROUND PEAK · 2026-07-31 11:15 UTC



09:45 UTC · 2026-07-31 · one point per minute, straight off the serving tier · 11:27 UTC

|                                                                                  |                                                                                                                           |                                                                                                       |                                                                                                                        |
|----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| <div>PEAK · MINUTE GRAIN</div> <div>19,882</div> <div>2026-07-31 11:15 UTC</div> | <div>PEAK · HOUR GRAIN</div> <div>19,882 / avg 9,911.6</div> <div>11:00 UTC — average is time-weighted, not sampled</div> | <div>CONSERVATION GATE</div> <div>1.000</div> <div>sum(opens) = sum(closes) = count() = 163,740</div> | <div>SLICE PEAKS MOVE</div> <div>1,100 @ 11:23</div> <div>SAMSUNG_HTML_TV — only 1,054 at the global peak minute</div> |
|----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|

# What went wrong, and what we did about it.

| THE CHALLENGE                             | WHAT IT LOOKED LIKE                                      | HOW WE SOLVED IT                                                                                                                                                                                                                 | RESULT                                          |
|-------------------------------------------|----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| A third of the day never closed           | 37,649 sessions with no End<br>25,403 with no Start      | An open session is published with a <b>provisional</b> end at its lease expiry. The next heartbeat re-derives it and the pipeline publishes the difference — <b>+1</b> at the old expiry, <b>-1</b> at the new one.              | 0 rebuilds<br>curve correct as of its watermark |
| Backgrounding almost never signalled      | only 6.51% of sessions carry both bg and fg              | The tuning day had 100% coverage and would have let us lean on bg/fg entirely. The lease was built as the <b>primary</b> exclusion signal instead, with the background event as a hard edge on top of it — not as a replacement. | held<br>no rule depended on bg arriving         |
| Liveness detection dropped Apple silently | 58.6% IPHONE<br>63.6% APPLE_TV<br>emit no heartbeat trio | Liveness qualifies on <b>any</b> event rather than the periodic trio. A trio-based rule returns a plausible number with most Apple traffic missing and raises nothing.                                                           | full coverage<br>21 platforms, none dropped     |
| Pause is invisible at the schema level    | 42 event names hidden under one event_type               | The playback axis is parsed from the event column, not event_type. Pause and error stop playback; play and resume restart it; error never terminates.                                                                            | 98,703<br>pause events correctly excluded       |
| A doubled curve passed every check we had | exactly 2x the peak, all invariants green                | Balance checks cannot detect a scalar multiple. Every stage now carries a <b>conservation check against its input</b> — sum ( opens ) in the deltas must equal count ( ) of intervals — and it throws with the ratio.            | ratio 1.000<br>on the unseen day                |
| Gates that only pass on data you know     | hardcoded tuning-day answers                             | Every gate was rewritten to be reference-free. Had the old ones survived they would have fired on <i>correct</i> unseen-day output and blocked the run — 19,882 is not 2,305.                                                    | 7 assertions<br>none contains a known number    |



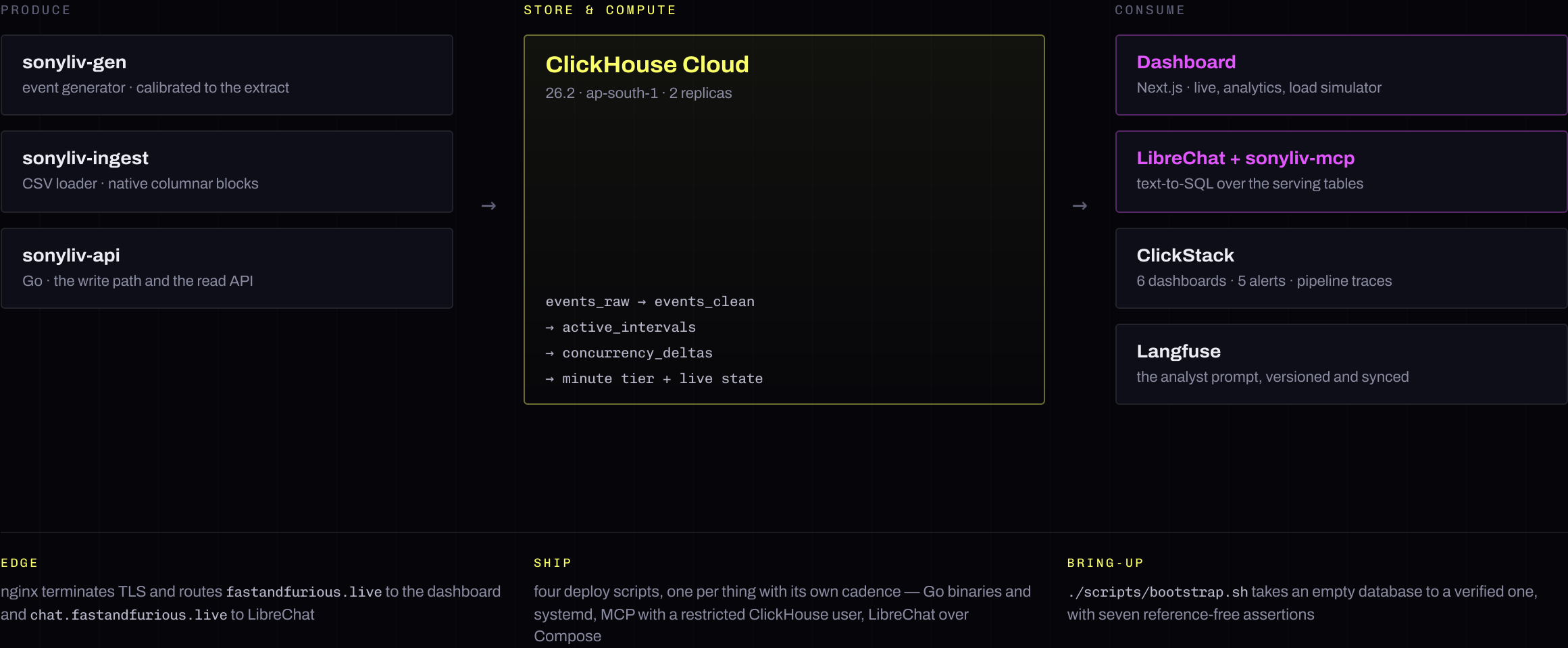
# The number that matters is not latency. It is **rows read**.

The brief says judges will look at what the queries read, not just how fast they return. Every row below is from the same service, on the same 7-million-event day.

| QUERY                                                                                   | ELAPSED | ROWS READ  | ANSWER           |
|-----------------------------------------------------------------------------------------|---------|------------|------------------|
| Day peak + time-weighted average                                                        | 5 ms    | 8,192      | 19,882 / 9,911.6 |
| Hour-grain peak + average, every hour                                                   | 5 ms    | 8,192      | 24 buckets       |
| Per-platform peak and each platform's own peak minute                                   | 7 ms    | 8,192      | 21 platforms     |
| Conservation gate over every delta row                                                  | 110 ms  | 677,016    | ratio 1.000      |
| Naive concurrency, recomputed from events<br>the thing the serving tier exists to avoid | 476 ms  | 13,945,916 | 24,069           |

|                                                                                                                                             |                                                                                                  |                                                                                   |                                                                                             |
|---------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| <div>VERSUS RECOMPUTING FROM EVENTS</div> <div>95× faster</div> <div>1,700× fewer rows — and the gap widens with every day of history</div> | <div>INGESTED</div> <div>7,000,000</div> <div>events, 0 rejected, 0 unjoinable content ids</div> | <div>ON DISK</div> <div>87.45 MiB</div> <div>raw + clean, 35.2× compression</div> | <div>SERVING P50</div> <div>7 ms</div> <div>from system.query_log, not a laptop proxy</div> |
|---------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|

# What is actually running.



# Ask it in **English**. It answers in SQL.

chat.fastandfurious.live

SCREENSHOT

chat.fastandfurious.live

Drop a PNG in here — see `pitch/README.md`. The instance is behind a login, so this frame is left empty rather than filled with something we did not capture.

IT CHANGED THE SCHEMA, NOT JUST THE DEMO

The honest reason `currency_minute_versions` exists is this integration. A flat, dense, one-table read surface is what makes text-to-SQL answerable. An LLM reliably writes `max(minute_peak) ... WHERE minute_start BETWEEN ...` It will **not** reliably write a day-anchored cumulative sum over millisecond boundaries.

| HOW IT IS WIRED   |                                                                                                                                              |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| sonyliv-mcp       | a Go MCP server on its own systemd unit, against a <b>restricted ClickHouse user</b> — the model cannot reach anything but the serving views |
| LiteLLM + Compose | model routing in front of LibreChat, deployed by <code>deploy-librechat.sh</code>                                                            |
| Langfuse          | holds the analyst prompt. It is <b>versioned and synced</b> , not pasted into a config — <code>langfuse-prompt-sync.sh</code>                |

# It observes **the pipeline itself.**

ClickStack – SonyLIV concurrency analytics

SCREENSHOT

### ClickStack dashboard

Drop a PNG in here — see `pitch/README.md`. ClickStack is reached through the Cloud console, so this frame is left empty rather than filled with something we did not capture.

SIX DASHBOARDS, CREATED THROUGH THE API

Not clicked together by hand — so they are diffable, reviewable and reproducible on another service. `apply.sh` is idempotent; `check-tiles.sh` runs every tile's SQL and reports pass/fail.

|                               |                       |
|-------------------------------|-----------------------|
| Live concurrency              | 10 s grain            |
| Corrected concurrency         | 1 min grain           |
| Pipeline observability        | lag · backlog · reads |
| Benchmark answers             | with query evidence   |
| Grouped viewers · drop alerts | per slice             |

FIVE ALERTS, AND ONE WE DELIBERATELY DO NOT FIRE

**Viewer drop** is measured against each slice's own trailing 15-minute median, not a global threshold — the brief's suggested LLM + ClickStack case. The alert carries the slice that moved, so the cause can be separated from the symptom.

**Not alerted:** “any late event”. Lateness is the baseline behaviour of this stream, and an alert that fires constantly is not an alert.

# Every requirement, and what we built for it.

| WHAT THE BRIEF REQUIRED                                  | WHAT WE BUILT                                                                                                                                                                | EVIDENCE, UNSEEN DAY                                                               |
|----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Correctness — foreground-only means foreground-only      | A five-clause predicate that excludes background, pause <i>and</i> silence, versioned in <code>policy.yaml</code> and stamped on every published answer.                     | <div>19,882 vs 24,069</div> <div>21.1% of the naive count was never watching</div> |
| Query performance — judges look at what you read         | A dense minute tier with no cumulative sum at read time, anchored per day so a window read never starts at <code>t = 0</code> .                                              | <div>5 ms · 8,192 rows</div> <div>1,700× fewer than recomputing</div>              |
| Update handling — absorb open sessions incrementally     | Provisional ends at lease expiry, then signed new–old corrections. No <code>ALTER UPDATE</code> , no rebuild, anywhere in the pipeline.                                      | <div>0 rebuilds</div> <div>34.7% of the day never closed</div>                     |
| Design quality — defend the trade-offs                   | Every switch carries its measured cost; every rejected alternative carries the number that rejected it. Conservation gates compare each stage against its input, not itself. | <div>ratio 1.000</div> <div><code>sum(opens) = count() = 163,740</code></div>      |
| ClickHouse as the primary datastore                      | Ingestion, modelling and every concurrency computation live in ClickHouse. Nothing is calculated in application code.                                                        | <div>7,000,000</div> <div>events, 0 rejected</div>                                 |
| Meaningfully integrate ClickStack / LibreChat / Langfuse | All three. ClickStack observes the pipeline, LibreChat answers over the serving tables, Langfuse versions the analyst prompt — and one of them changed the schema.           | <div>6 dashboards</div> <div>5 alerts · 1 MCP server</div>                         |
| The unseen day — no pipeline evidence, no credit         | Full-scan and incremental are the same file, so the oracle cannot drift from the path it validates. Every figure in this deck came back out of the service.                  | <div>query_log</div> <div>behind every number here</div>                           |



×



# Thank you.

Counting real viewers, not ghosts.

● [fastandfurious.live](#) — the concurrency curve, live

● [chat.fastandfurious.live](#) — ask it in plain English

CLICK-A-THON 2026 · BENGALURU · SONYLIV TRACK · MIT LICENSED, PUBLIC REPOSITORY · 19,882 CONCURRENT, VERIFIED

